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Project Report

For

Kheti Saathi: Crop Recommendation and Plant Disease Detection Application

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ABSTRACT

Agriculture plays a vital role in the economy of many countries, particularly India, where a large portion of the population depends on farming for their livelihood. However, farmers often face multiple challenges such as plant diseases, poor soil conditions, unpredictable climate changes, and lack of proper crop planning. These problems lead to reduced crop yield and financial losses for farmers.

Kheti Saathi is an intelligent agricultural support system designed to assist farmers in making better farming decisions using modern technologies such as machine learning and artificial intelligence. The system provides two major functionalities: plant disease detection and crop recommendation.

The plant disease detection module allows farmers to capture images of plant leaves using their smartphone cameras. The system analyzes the image using a trained machine learning model to detect whether the plant is healthy or infected with a disease. After detection, the system provides detailed information about the disease along with recommended treatments and preventive measures.

In addition to disease detection, the system also includes a crop recommendation module. This module suggests suitable crops for cultivation based on environmental and soil parameters such as temperature, humidity, soil nutrients, rainfall, and pH levels. By analyzing these factors, the system helps farmers choose the most suitable crop for their land conditions.

The Kheti Saathi application is developed using Flutter for mobile application development and integrates machine learning models for intelligent predictions. The main objective of the system is to empower farmers with technological tools that improve crop health monitoring and assist in better crop selection decisions.

Chapter 1 – Introduction

1.1 Background

Agriculture is one of the most important sectors of the economy, especially in countries like India where a large portion of the population depends on farming for their livelihood. However, agricultural productivity is often affected by various challenges such as plant diseases, improper crop selection, climate variations, and lack of technical knowledge. Among these challenges, plant diseases are a major factor that can significantly reduce crop yield and quality if not identified and treated at an early stage.

Traditionally, farmers identify plant diseases by observing symptoms on leaves, stems, or fruits. However, many diseases have similar symptoms, making it difficult for farmers to correctly diagnose the problem. Incorrect identification may lead to the use of inappropriate pesticides, which can damage crops and increase farming costs. In addition, farmers often face difficulty in selecting the most suitable crop for cultivation based on soil nutrients and environmental conditions, which may result in poor agricultural productivity.

With the advancement of artificial intelligence and machine learning technologies, it has become possible to develop smart systems that assist farmers in solving these problems. Image

processing techniques can analyze plant leaf images and detect diseases automatically, while machine learning models can recommend suitable crops based on soil and climate parameters.

The **Kheti Saathi** system is developed as an intelligent agricultural support application that integrates plant disease detection and crop recommendation to help farmers make better decisions and improve crop productivity.

1.2 Need for the System

The development of the **Kheti Saathi** system is necessary due to several challenges faced by farmers in traditional agricultural practices. One of the major problems is the lack of proper knowledge about plant diseases. Farmers often find it difficult to correctly identify diseases based only on visible symptoms on leaves or stems. As a result, they may apply incorrect treatment methods or pesticides, which can worsen the condition of the crop instead of improving it.

Another issue is the limited availability of agricultural experts in rural areas. Farmers may need to travel long distances to consult agricultural specialists or visit research centers for guidance. This process takes time, and during this delay the disease may continue to spread across the crop field, causing greater damage.

Crop selection is another important challenge in agriculture. The productivity of crops depends on several environmental and soil-related factors such as soil nutrients, temperature, humidity, rainfall, and soil pH levels. However, many farmers do not have access to tools or systems that can analyze these factors and recommend the most suitable crop for cultivation. As a result, farmers may grow crops that are not suitable for their soil or climate conditions, leading to poor yield and financial loss.

In addition, farmers often lack a reliable and easily accessible platform that can provide both disease diagnosis and crop recommendations in one place. Therefore, there is a strong need for an intelligent system that can assist farmers in identifying plant diseases and recommending suitable crops based on environmental and soil parameters.

The **Kheti Saathi** system addresses this need by providing a mobile-based solution that uses artificial intelligence and machine learning technologies to support farmers in making better agricultural decisions and improving crop productivity.

1.3 Scope of the Project

The **Kheti Saathi** project focuses on developing a mobile application that helps farmers detect plant diseases using machine learning and image processing techniques. The system allows farmers to capture images of plant leaves using the smartphone camera, which are then analyzed by a trained machine learning model to determine whether the plant is healthy or affected by a disease.

After detecting the disease, the system provides information about the disease along with possible treatment methods and preventive measures. In addition, the system includes a crop recommendation feature that suggests suitable crops based on soil nutrients and environmental conditions such as temperature, humidity, rainfall, and soil pH.

The scope of the project includes developing a user-friendly mobile application, integrating machine learning models for disease detection and crop recommendation, and implementing image processing techniques. In the future, the system can be expanded by adding features such as weather-based farming suggestions, fertilizer recommendations, soil analysis, and market price information.

1.4 Research Relevance

Recent research in the field of smart agriculture highlights the growing importance of artificial intelligence and machine learning for improving crop monitoring and agricultural productivity. Several studies have shown that deep learning techniques, particularly convolutional neural networks (CNNs), can effectively detect plant diseases from leaf images.

For example, a study by **Mohanty, Hughes, and Salathé (2016)** trained a deep CNN model using more than 54,000 plant leaf images from the PlantVillage dataset and achieved an accuracy of **about 99.35% in identifying 26 different plant diseases across 14 crop species**. This study demonstrated the strong potential of image-based disease detection using deep learning.

Further research also shows that CNN-based models are highly effective for image recognition tasks in agriculture because they automatically learn visual features such as color patterns, leaf textures, and disease spots. Many recent studies confirm that deep learning approaches provide reliable and rapid diagnosis of plant diseases, helping farmers detect crop problems earlier and reduce losses.

In addition to plant disease detection, machine learning is also widely used for crop recommendation systems. These systems analyze soil nutrients, environmental conditions, and climatic factors to suggest suitable crops for cultivation. Research indicates that machine learning models can evaluate parameters such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, rainfall, and soil pH to recommend the most suitable crops for a particular region or field.

Recent studies also demonstrate that crop recommendation models based on advanced machine learning algorithms, such as gradient boosting or random forest, can achieve very high prediction accuracy when trained on agricultural datasets containing soil and climate information.

Therefore, the Kheti Saathi project aligns closely with current research trends in precision agriculture. By combining plant disease detection using image analysis with crop recommendation based on soil and environmental parameters, the system contributes to the development of intelligent farming solutions that support farmers in making better agricultural decisions and improving crop productivity.

1.5 Objectives

The main objectives of the Kheti Saathi project are:

- To develop a mobile application that can detect plant diseases using Machine learning and image processing techniques.

- To enable farmers to capture or upload plant leaf images for automatic disease Identification.
- To provide information about detected diseases, including causes, treatment methods, and preventive measures.
- To develop a crop recommendation system that suggests suitable crops based on soil nutrients and environmental conditions.
- To help farmers make better agricultural decisions and improve crop productivity using modern technology.

Chapter 2 – Problem Statement

Farmers often face difficulties in maintaining crop health and selecting suitable crops for cultivation due to a lack of proper technological support and accurate information. One of the major problems in agriculture is the late identification of plant diseases. Many diseases appear with similar symptoms such as yellow spots, leaf discoloration, or damage to plant tissues, which makes it difficult for farmers to correctly identify the problem through visual inspection alone.

Due to incorrect disease identification, farmers may use inappropriate pesticides or treatment methods, which can further damage the crops and increase production costs. In many rural areas, farmers also have limited access to agricultural experts who can provide timely guidance. This delay in diagnosis allows the disease to spread further across the field, resulting in reduced crop yield.

Another significant problem is the lack of scientific guidance for crop selection. The success of crop cultivation depends on various environmental and soil factors such as soil nutrients, rainfall, humidity, temperature, and soil pH. However, farmers often choose crops based on traditional practices rather than analyzing these important factors. As a result, unsuitable crops may be cultivated in unfavorable conditions, which can lead to low productivity and financial losses.

Furthermore, farmers generally do not have a single platform that provides both plant disease detection and crop recommendation in an easy and accessible manner. Therefore, there is a need for a smart system that can assist farmers in identifying plant diseases quickly and recommending suitable crops based on soil and environmental conditions.

The Kheti Saathi system is proposed to address these challenges by using machine learning and image processing techniques to detect plant diseases and recommend appropriate crops through a mobile-based application.

Chapter 3 – Proposed Solution

To address the problems faced by farmers in identifying plant diseases and selecting suitable crops, the Kheti Saathi system proposes an intelligent mobile-based solution.

The application allows farmers to capture images of plant leaves using their smartphone cameras. These images are analyzed using a trained machine learning model that detects

whether the plant is healthy or affected by a disease. After identifying the disease, the system provides detailed information about the disease along with recommended treatment methods and preventive measures.

In addition to disease detection, the system also includes a crop recommendation feature. This feature analyzes important parameters such as soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, rainfall, and soil pH levels to recommend the most suitable crops for cultivation. By combining both disease detection and crop recommendation, the system acts as a digital assistant for farmers and helps them improve crop productivity.

3.1 Working of the System

The working of the Kheti Saathi system involves several steps.

First, the user opens the mobile application and selects the desired feature such as plant disease detection or crop recommendation.

For disease detection, the farmer captures or uploads an image of a plant leaf using the smartphone camera. The captured image is then processed using image preprocessing techniques such as resizing and normalization. After preprocessing, the image is passed to a trained machine learning model which analyzes the visual features of the leaf and predicts whether the plant is healthy or infected by a disease.

Once the disease is detected, the system displays the name of the disease along with relevant information such as causes, treatment suggestions, and preventive measures. This helps farmers take appropriate actions to protect their crops.

For crop recommendation, the farmer enters environmental and soil parameters such as nitrogen level, phosphorus level, potassium level, temperature, humidity, rainfall, and soil pH. The machine learning model processes these inputs and predicts the most suitable crop that can be grown under those conditions.

The results are then displayed on the application interface, helping farmers make better agricultural decisions.

3.2 Key Features

The Kheti Saathi system provides several important features that make it useful for farmers.

1. **Plant Disease Detection:** The application can identify plant diseases using leaf images captured by the smartphone camera.
2. **Crop Recommendation:** The system suggests suitable crops based on soil nutrients and environmental conditions.
3. **User-Friendly Interface:** The mobile application is designed to be simple and easy to use so that farmers can operate it without technical knowledge.
4. **Treatment Suggestions:** The system provides recommendations for disease treatment and preventive measures.

5. **Real-Time Analysis:** The system quickly analyzes the input data and provides instant results to the user.

3.3 Advantages of the Proposed System

The proposed system offers several advantages in improving agricultural practices.

One of the major advantages is the early detection of plant diseases, which helps farmers take timely actions to prevent crop damage. Early diagnosis can significantly reduce crop loss and improve yield.

Another advantage is that the system provides scientific crop recommendations based on soil and environmental parameters. This helps farmers select crops that are most suitable for their land conditions.

The system also reduces the dependency on agricultural experts by providing quick and reliable guidance through a mobile application. Farmers can access the system anytime using their smartphones.

Additionally, the application integrates multiple agricultural services in one platform, making it easier for farmers to monitor crop health and plan their farming activities effectively.

Overall, the Kheti Saathi system contributes to the development of smart agriculture by combining artificial intelligence with mobile technology to support farmers and improve agricultural productivity.

Chapter 4 – Tools and Technologies

The development of the Kheti Saathi system requires the use of various software tools and technologies to implement plant disease detection and crop recommendation features. These technologies help in building a reliable and efficient system that can analyze data, process images, and provide accurate results to the users. The system mainly uses mobile application development tools, machine learning frameworks, and databases to perform its operations.

4.1 Frontend Technology

The frontend of the system is developed using the **Flutter framework**, which is an open-source UI toolkit developed by Google. Flutter allows developers to create cross-platform mobile applications using a single codebase. It provides a rich set of widgets that help in designing an interactive and user-friendly interface.

In the Kheti Saathi application, Flutter is used to design screens where users can capture plant leaf images, enter soil parameters, and view disease detection results or crop recommendations. The interface is designed to be simple and easy to use so that farmers with minimal technical knowledge can operate the application without difficulty.

4.2 Backend Technology

The backend of the system handles the processing of user inputs, communication with machine learning models, and retrieval of results. It manages the logical operations required to process the images and input parameters provided by the user.

Machine learning models are integrated with the backend to analyze plant leaf images and environmental data. The backend ensures that the application processes the inputs efficiently and returns accurate predictions to the user.

4.3 Database

A database is used to store important information related to plant diseases, treatment suggestions, and crop recommendations. The database helps in organizing and retrieving data quickly whenever the system needs to display information to the user.

The database may contain details such as disease names, symptoms, causes, treatment methods, and preventive measures. It also stores crop information related to soil nutrients and environmental conditions required for crop growth.

4.4 Machine Learning Model

Machine learning models play a crucial role in the Kheti Saathi system. For plant disease detection, a Convolutional Neural Network (CNN) model is used because it is highly effective in image classification tasks. The model is trained using a dataset of plant leaf images containing both healthy and diseased samples.

For crop recommendation, a machine learning algorithm such as Decision Tree, Random Forest, or other classification models can be used to analyze environmental parameters and suggest suitable crops.

These models help the system make intelligent predictions based on the input data provided by the user.

4.5 Image Processing Techniques

Image processing techniques are used to prepare plant leaf images before sending them to the machine learning model. These techniques help improve the quality and consistency of the input data.

Common preprocessing steps include image resizing, normalization, and noise removal. These steps ensure that the images are in a suitable format for the machine learning model to analyze.

By applying image processing techniques, the system can detect important visual patterns in plant leaves such as spots, discoloration, or texture changes that indicate plant diseases.

Chapter 5 – Methodology

The methodology describes the step-by-step process followed in the development of the Kheti Saathi system. It explains how the system was planned, designed, implemented, and tested to ensure that it performs plant disease detection and crop recommendation effectively. A

systematic approach was followed so that each stage of development could be completed in an organized and efficient manner.

The overall methodology includes requirement analysis, data collection, system design, implementation, testing, and deployment. Each stage contributes to building a reliable and functional system.

5.1 Requirement Analysis

The first step in the development process is identifying the requirements of the system. During this phase, the problems faced by farmers were studied to understand the need for plant disease detection and crop recommendation. Farmers often find it difficult to identify plant diseases accurately and select suitable crops based on soil and environmental conditions.

Based on these observations, the system requirements were defined. The application should allow farmers to capture plant leaf images, detect diseases using machine learning, and recommend crops based on environmental parameters. The system should also provide treatment suggestions and preventive measures.

In addition to functional requirements, non-functional requirements such as ease of use, quick response time, and compatibility with mobile devices were also considered.

5.2 Data Collection

Data collection is an important step for training machine learning models. For plant disease detection, a dataset containing images of both healthy and diseased plant leaves was collected. These images help train the model to recognize visual patterns associated with different diseases.

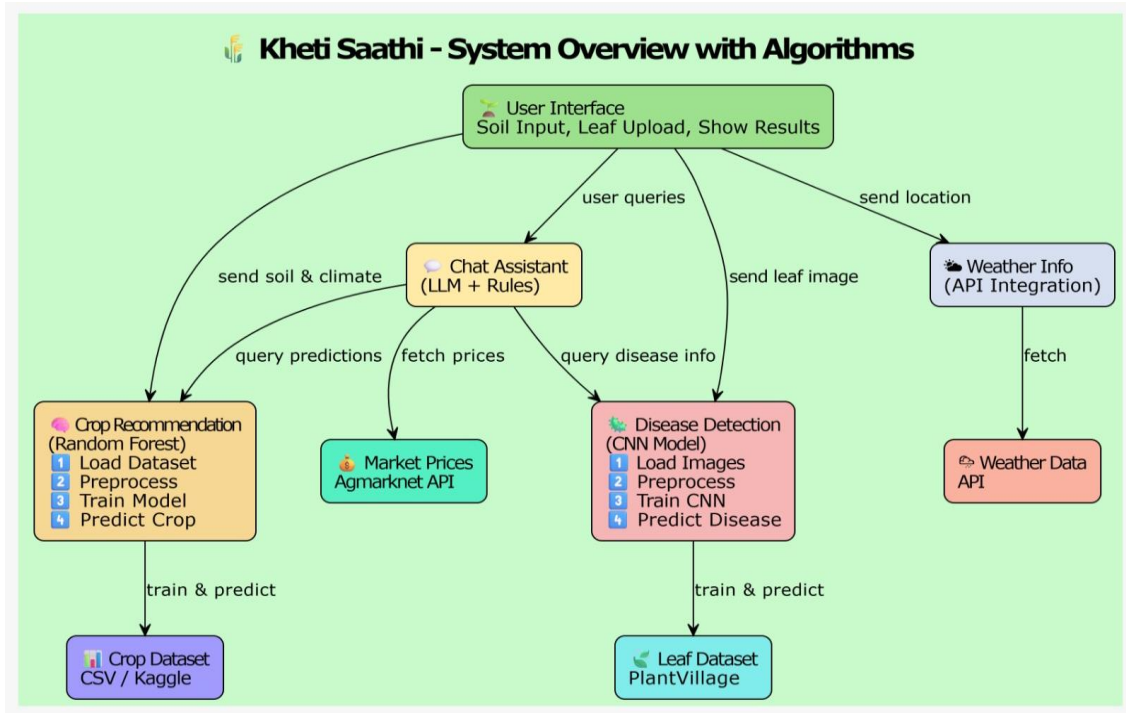
For the crop recommendation module, a dataset containing soil nutrients such as nitrogen (N), phosphorus (P), and potassium (K), along with environmental parameters like temperature, humidity, rainfall, and soil pH levels, was collected. This dataset allows the machine learning model to understand the relationship between environmental conditions and suitable crops.

5.3 System Design

In this stage, the architecture of the Kheti Saathi system was designed. The system is divided into different modules such as the user interface module, disease detection module, crop recommendation module, and database module.

The user interface allows farmers to interact with the application by capturing images or entering soil and environmental parameters. The disease detection module processes leaf images using image processing and machine learning techniques. The crop recommendation module analyzes input parameters and predicts the most suitable crop for cultivation.

This modular design helps improve system organization and allows easier updates or improvements in the future.



5.4 Implementation

During the implementation phase, the mobile application was developed using the Flutter framework. Flutter was chosen because it supports cross-platform mobile application development and provides an interactive user interface.

The machine learning models for plant disease detection and crop recommendation were integrated into the application. Image preprocessing techniques such as resizing and normalization were applied to leaf images before sending them to the machine learning model for prediction.

The application interface was designed to ensure that farmers can easily capture images, enter parameters, and view results without difficulty.

5.5 Testing

Testing was carried out to ensure that the system works correctly and produces accurate results. The disease detection model was tested with multiple plant leaf images to evaluate its prediction accuracy.

Similarly, the crop recommendation system was tested using different combinations of soil and environmental parameters. The results were verified to ensure that the recommended crops match the given conditions.

Usability testing was also performed to confirm that the mobile application interface is easy to use and functions properly.

5.6 Deployment

After successful testing, the system is ready for deployment. The mobile application can be installed on smartphones so that farmers can easily use it in their fields.

The system can also be updated periodically by adding new datasets and improving machine learning models. This will help increase the accuracy of disease detection and expand the crop recommendation system to support more crops and agricultural conditions.

Chapter 6 – System Architecture and Modules

The Kheti Saathi system is designed with a modular architecture that allows different components of the system to work together efficiently. The architecture mainly consists of the mobile application interface, machine learning models for prediction, and a database for storing information related to plant diseases and crop recommendations. Each component of the system performs a specific function, making the application easier to manage, maintain, and expand in the future.

The user interacts with the system through the mobile application interface. The application collects inputs such as plant leaf images or environmental parameters from the user. These inputs are processed using machine learning models to generate predictions. The results are then displayed on the user interface along with useful information and recommendations.

To organize the working of the system effectively, the Kheti Saathi application is divided into several modules.

6.1 User Interface Module

The user interface module is responsible for providing interaction between the user and the application. It allows farmers to easily navigate through the application and access its features.

This module provides options for capturing plant leaf images using the smartphone camera, uploading images from the gallery, and entering environmental parameters required for crop recommendation. It also displays the results generated by the system, including disease detection results and crop recommendations.

The interface is designed to be simple and user-friendly so that farmers with minimal technical knowledge can easily use the application.

6.2 Image Capture Module

The image capture module allows the user to capture images of plant leaves using the smartphone camera or upload images from the device gallery. This module ensures that the image is properly captured and prepared for processing.

Once the image is captured, it is sent to the image processing stage where it undergoes preprocessing steps such as resizing and normalization before being analyzed by the machine learning model.

6.3 Disease Detection Module

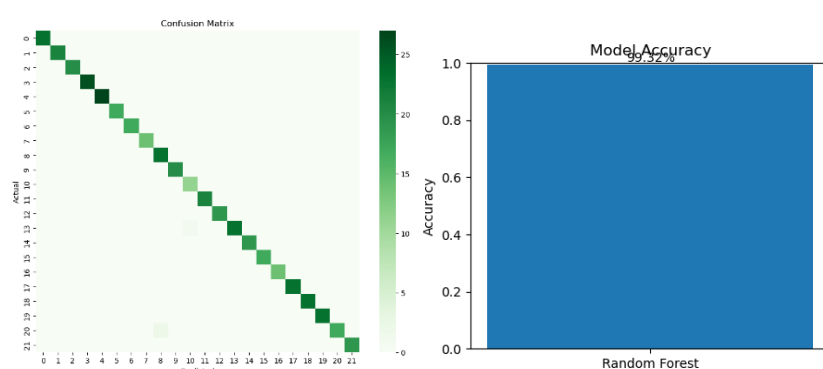
The disease detection module is responsible for identifying plant diseases from the captured leaf images. This module uses a trained machine learning model, typically a convolutional neural network (CNN), to analyze the visual features of the leaf image.

The model examines patterns such as spots, discoloration, and texture changes on the leaf surface to determine whether the plant is healthy or affected by a disease. After processing the image, the system displays the predicted disease name along with information about the disease and recommended treatment methods.

6.4 Crop Recommendation Module

The crop recommendation module helps farmers select suitable crops based on environmental and soil parameters. The user enters parameters such as nitrogen, phosphorus, potassium levels, temperature, humidity, rainfall, and soil pH.

The machine learning model analyzes these parameters and predicts the most suitable crop for cultivation. This module helps farmers make informed decisions about crop selection and improves agricultural productivity.



6.5 Database Module

The database module stores information related to plant diseases, treatment methods, preventive measures, and crop recommendations. It helps organize and manage all the necessary data used by the application.

When a disease is detected or a crop is recommended, the system retrieves relevant information from the database and displays it to the user. The database can also be updated with new diseases, treatments, or crop information to improve the system over time.

	A	B	C	D	E	F	G	H
1	N	P	K	temperatu	humidity	ph	rainfall	label
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice

Chapter 7 – Algorithms Used

The Kheti Saathi system uses machine learning algorithms and image processing techniques to perform plant disease detection and crop recommendation. These algorithms help the system analyze input data such as plant leaf images and environmental parameters to generate accurate predictions. The use of intelligent algorithms allows the system to assist farmers in making better agricultural decisions.

7.1 Image Processing for Leaf Analysis

Image processing techniques are used to prepare plant leaf images before they are analyzed by the machine learning model. When a user captures or uploads an image of a plant leaf, the image may contain noise, different lighting conditions, or unnecessary background elements.

To improve the quality of the image, preprocessing steps such as resizing, normalization, and noise reduction are applied. These steps ensure that all images are in a consistent format before being processed by the machine learning model.

Image preprocessing helps the model focus on important visual features of the leaf, such as spots, discoloration, or texture changes, which are useful for identifying plant diseases.

7.2 Convolutional Neural Network (CNN)

For plant disease detection, the system uses a Convolutional Neural Network (CNN) model. CNN is a deep learning algorithm that is highly effective for image classification tasks.

The CNN model analyzes the input leaf image by extracting different visual features through multiple layers. These layers detect patterns such as edges, shapes, and textures present in the image. By learning these patterns during training, the model can identify different plant diseases based on the appearance of the leaf.

Once the model processes the image, it predicts whether the plant is healthy or affected by a specific disease.

1. Convolution Operation

$$F(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n) K(i-m, j-n)$$

Where:

- I = Input image
- K = Kernel / filter
- $F(i,j)$ = Feature map output

7.3 Crop Recommendation Algorithm

For the crop recommendation feature, a machine learning classification algorithm is used to analyze environmental and soil parameters. The system takes inputs such as nitrogen, phosphorus, potassium levels, temperature, humidity, rainfall, and soil pH.

These parameters are processed by the machine learning model to determine the most suitable crop that can be grown under the given conditions. Algorithms such as Decision Tree, Random Forest, or other classification models can be used for this purpose.

The model compares the input parameters with patterns learned during training and predicts the crop that has the highest probability of growing successfully.

7.4 Prediction Process

The prediction process begins when the user provides input to the system. For disease detection, the input is a plant leaf image, while for crop recommendation the input consists of environmental and soil parameters.

The system processes these inputs using the appropriate algorithm and generates predictions. The final results are then displayed to the user through the mobile application along with additional information such as treatment suggestions or recommended crops.

By combining image processing, deep learning, and machine learning techniques, the Kheti Saathi system provides reliable and intelligent agricultural support to farmers.

Chapter 8 – Implementation and Working

The implementation of the Kheti Saathi system involves the integration of a mobile application with machine learning models to provide plant disease detection and crop recommendation services. The system is designed to be simple and user-friendly so that farmers can easily use it without requiring advanced technical knowledge.

The application is developed using modern mobile development tools and machine learning frameworks. The main goal of the implementation is to ensure that the system can efficiently process user inputs and provide accurate results.

8.1 Starting the Application

When the user opens the Kheti Saathi mobile application, the main interface is displayed. The home screen provides options for different functionalities such as Plant Disease Detection and Crop Recommendation. The user can select the desired feature based on their requirement.

The interface is designed to be simple and clear so that farmers can easily navigate through the application.

8.2 Plant Disease Detection Process

For plant disease detection, the user selects the disease detection option in the application. The user can then capture an image of a plant leaf using the smartphone camera or upload an existing image from the device gallery.

After the image is selected, the application performs image preprocessing steps such as resizing and normalization to prepare the image for analysis. The processed image is then passed to the trained machine learning model, which analyzes the visual features of the leaf.

The model predicts whether the plant is healthy or affected by a disease. The result is displayed on the screen along with information about the detected disease, possible causes, and suggested treatment methods.

8.3 Crop Recommendation Process

For crop recommendation, the user enters the required environmental and soil parameters into the application. These parameters include nitrogen, phosphorus, potassium levels, temperature, humidity, rainfall, and soil pH.

The system processes these inputs using a trained machine learning model that analyzes the parameters and predicts the most suitable crop for cultivation. The recommended crop is then displayed to the user through the application interface.

This feature helps farmers select crops that are suitable for their soil and environmental conditions.

8.4 Displaying Results

After the prediction process is completed, the results are displayed in a clear and understandable format. For disease detection, the application shows the disease name along with relevant information such as symptoms, treatment suggestions, and preventive measures.

For crop recommendation, the application displays the most suitable crop along with supporting information that helps farmers understand the recommendation.

8.5 Error Handling

The system also includes basic error handling to manage situations such as incorrect inputs or poor image quality. If the image captured by the user is unclear or unsuitable for analysis, the system may prompt the user to capture a clearer image.

Similarly, if invalid or incomplete environmental parameters are entered for crop recommendation, the system requests the user to provide the correct values.

8.6 Overall Working Flow

The overall working of the system follows a simple flow. The user provides input to the application either by capturing a plant leaf image or by entering environmental parameters. The system processes this input using machine learning algorithms and generates predictions. Finally, the application displays the results along with useful information to help the user make better agricultural decisions.

Through this implementation, the Kheti Saathi system provides farmers with an easy-to-use digital tool that supports plant disease detection and crop selection.

Chapter 9 – Applications and Advantages

The Kheti Saathi system is designed to support farmers in improving crop health management and making better agricultural decisions. By using machine learning and mobile technology, the system can be applied in various areas of agriculture and related fields. The integration of plant disease detection and crop recommendation makes the system useful for both farmers and agricultural researchers.

9.1 Applications

The Kheti Saathi application can be used in several practical agricultural scenarios.

One of the main applications is plant disease detection. Farmers can use the application to quickly identify diseases affecting their crops by capturing images of plant leaves. This helps them take timely actions to control the disease and prevent further crop damage.

Another important application is crop recommendation. Farmers can use the system to determine which crops are most suitable for their land based on soil nutrients and environmental conditions. This helps improve crop productivity and reduce the chances of crop failure.

The system can also be useful for agricultural research and education. Agricultural students and researchers can use the application to study plant diseases and understand how environmental factors affect crop growth.

Additionally, the system can support smart farming practices by providing digital tools that assist farmers in managing crop health more efficiently.

9.2 Advantages

The Kheti Saathi system provides several advantages that help farmers improve their agricultural practices:

1. **Early Disease Detection:** Helps farmers identify plant diseases at an early stage, reducing crop damage.
2. **Accurate Crop Recommendation:** Suggests suitable crops based on soil nutrients and environmental conditions.
3. **Improved Crop Productivity:** Helps farmers choose the right crops and treatments, leading to better yield.
4. **Time Saving:** Farmers can quickly detect diseases and get recommendations without visiting agricultural experts.
5. **User-Friendly Application:** The mobile application is simple and easy to use for farmers.
6. **Accessible Anywhere:** Farmers can use the application directly on their smartphones in the field.
7. **Reduced Crop Loss:** Early diagnosis and proper crop selection help minimize agricultural losses.
8. **Support for Smart Farming:** Integrates modern technologies like machine learning and image processing in agriculture.

Chapter 10 – SWOT Analysis

SWOT analysis is used to evaluate the Strengths, Weaknesses, Opportunities, and Threats of the Kheti Saathi system. It helps in understanding the advantages of the system as well as the possible challenges that may arise during its implementation and usage.

10.1 Strengths

1. The system provides early detection of plant diseases using machine learning and image processing techniques.
2. It offers crop recommendations based on soil nutrients and environmental conditions, helping farmers choose suitable crops.
3. The application is user-friendly and easy to operate, making it accessible even for farmers with limited technical knowledge.
4. It integrates multiple agricultural services in a single mobile application.
5. The system helps farmers reduce crop losses and improve productivity.

10.2 Weaknesses

1. The accuracy of disease detection depends on the quality of images captured by the user.
2. The system requires proper datasets for training the machine learning models.
3. Farmers without smartphones or internet access may not be able to use the application easily.
4. Limited dataset coverage may restrict the system to a specific number of crops and diseases.

10.3 Opportunities

1. The system can be expanded to support more crops and plant diseases in the future.
2. It can be integrated with weather forecasting systems for better agricultural recommendations.
3. The application can be improved by adding multi-language support for farmers from different regions.
4. Integration with IoT sensors and smart farming technologies can further enhance the system.

10.4 Threats

1. Incorrect predictions may occur if the input data is inaccurate or incomplete.
2. Rapid changes in environmental conditions may affect crop recommendation accuracy.
3. Farmers may initially hesitate to trust automated systems for agricultural decisions.
4. Continuous system updates and maintenance are required to keep the data and models accurate.

Chapter 11 – System Requirements

The Kheti Saathi system requires certain hardware and software components for development and proper functioning. These requirements ensure that the application can run smoothly and

perform tasks such as image processing, disease detection, and crop recommendation efficiently.

11.1 Hardware Requirements

The hardware requirements refer to the physical devices needed to run the application.

1. **Smartphone with Camera:** Required to capture plant leaf images for disease detection.
2. **Processor:** Minimum dual-core processor for smooth application performance.
3. **RAM:** At least 4 GB RAM recommended for efficient processing.
4. **Storage:** Sufficient storage space to install and run the mobile application.
5. **Internet Connectivity:** Required for accessing online features and updates.

11.2 Software Requirements

Software requirements include the programming tools, frameworks, and platforms used to develop and run the system.

1. **Operating System:** Android operating system for running the mobile application.
2. **Frontend Development:** Flutter framework for designing the mobile user interface.
3. **Programming Languages:** Dart for Flutter development and Python for machine learning model development.
4. **Machine Learning Framework:** TensorFlow or PyTorch for training the disease detection model.
5. **Database:** A database system to store disease information and crop data.

11.3 Network Requirements

Network requirements ensure that the system can communicate and retrieve necessary data when required.

1. **Internet Connection:** Required for updating datasets and accessing additional resources.
2. **Mobile Network or Wi-Fi:** Enables communication between the mobile application and backend services if needed.
3. **Cloud Support (Optional):** Cloud platforms can be used for storing models and datasets for better scalability.

Chapter 12 – Deployment and Maintenance

After the development and testing phases, the Kheti Saathi system needs to be deployed so that users can start using the application. Deployment involves making the mobile application available on smartphones and ensuring that all system components work correctly in the real environment. Proper maintenance is also important to keep the system updated and functioning efficiently over time.

12.1 Deployment Strategy

The Kheti Saathi mobile application can be deployed on Android smartphones so that farmers can easily access it in the field. The application can be installed directly on mobile devices or distributed through platforms such as the Google Play Store.

Before deployment, the system is tested to ensure that all features such as disease detection and crop recommendation work correctly. The machine learning models are integrated with the mobile application so that predictions can be generated quickly when users provide inputs.

Proper documentation and instructions can also be provided to help users understand how to use the application effectively.

12.2 Maintenance

Maintenance is necessary to ensure that the system continues to work correctly after deployment. Over time, new plant diseases may appear and new agricultural data may become available. Therefore, the system should be updated regularly with new datasets and improved machine learning models.

Maintenance also includes fixing bugs, improving application performance, and adding new features based on user feedback. Regular updates help maintain the accuracy and reliability of the system.

12.3 Monitoring

Monitoring is an important part of maintaining the system. It involves checking whether the application is working properly and whether users are receiving accurate results.

User feedback can be collected to identify any issues with disease detection or crop recommendation. If errors or inaccuracies are reported, the system can be improved by updating the machine learning models or adjusting the data used for predictions.

Continuous monitoring ensures that the Kheti Saathi system remains reliable and useful for farmers.

Chapter 13 – Conclusion

The Kheti Saathi system provides an intelligent and practical solution to some of the major challenges faced in modern agriculture. Farmers often struggle with identifying plant diseases and selecting suitable crops due to limited access to expert guidance and technological tools. The developed system addresses these problems by integrating machine learning and image processing techniques into a mobile application that is simple and easy to use.

The application allows farmers to detect plant diseases by capturing images of plant leaves and analyzing them using a trained machine learning model. The system then provides information about the disease along with possible treatment methods and preventive measures. In addition, the crop recommendation feature helps farmers choose suitable crops based on soil nutrients and environmental conditions such as temperature, humidity, rainfall, and soil pH levels.

By combining these features in a single platform, the Kheti Saathi system helps farmers make better agricultural decisions and improve crop productivity. The use of modern technologies makes the system efficient, accessible, and capable of supporting smart farming practices.

In the future, the system can be enhanced by adding more crop varieties, supporting multiple languages, and integrating additional features such as weather-based recommendations and pest detection.

Overall, the Kheti Saathi system demonstrates how artificial intelligence can be effectively used to support farmers and contribute to the advancement of agriculture.

Chapter 14 – References

Below are recent research papers and sources related to plant disease detection and crop recommendation using machine learning, which support the concept of the Kheti Saathi project.

1. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016).

Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science*.

This research demonstrated that deep learning models trained on large datasets of plant leaf images can accurately identify plant diseases across multiple crop species.

2. Kim, T. et al. (2024).

ANFIS Fuzzy Convolutional Neural Network Model for Leaf Disease Detection. *Frontiers in Plant Science*.

This research proposes an advanced CNN-based approach that improves plant leaf disease detection accuracy by analyzing visual patterns in plant leaves.

3. Shafay, M. et al. (2025).

Recent Advances in Plant Disease Detection Using Artificial Intelligence.

The study highlights how AI-based disease detection systems can improve agricultural productivity and support sustainable farming practices.

4. Fu, Y. et al. (2024).

A Lightweight CNN Model for Pepper Leaf Disease Recognition.

This paper introduces a CNN model designed to detect plant diseases efficiently and demonstrates the effectiveness of deep learning in crop monitoring.

5. Prity, F. S. et al. (2024).

A Machine Learning Approach to Crop Recommendation Systems.

The research proposes a crop recommendation system that uses soil properties, climatic conditions, and historical agricultural data to suggest suitable crops.

6. Shastri, S. et al. (2025). **Advancing Crop Recommendation Systems with Supervised Machine Learning.**

The research demonstrates how machine learning algorithms can analyze agricultural data to recommend suitable crops and improve yield prediction.

7. Kanakala, S. & Ningappa, S. (2025).

Detection and Classification of Diseases in Multi-Crop Leaves using CNN and LSTM Models.

This study demonstrates that deep learning models can detect multiple plant diseases with high accuracy using large datasets of leaf images.